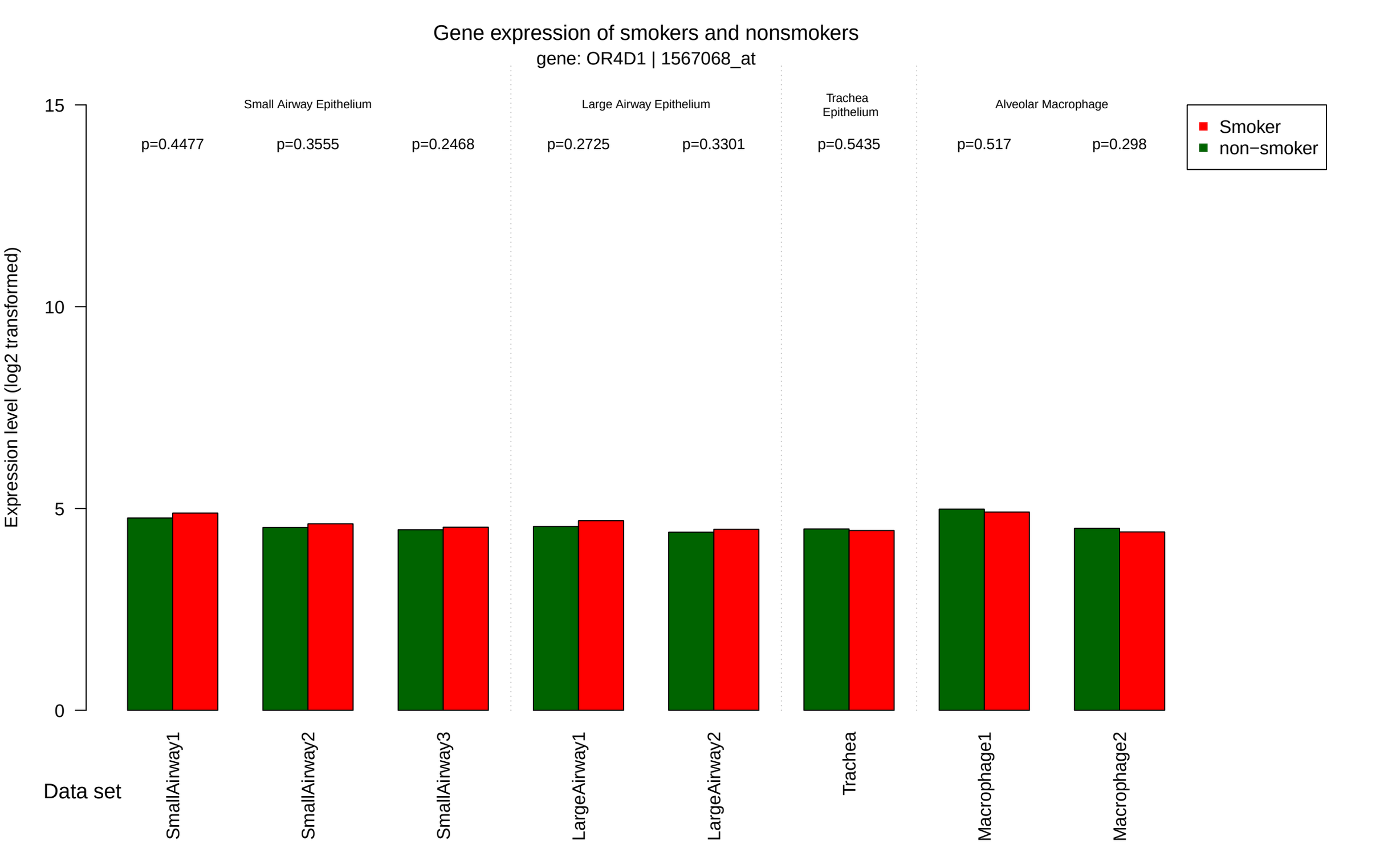
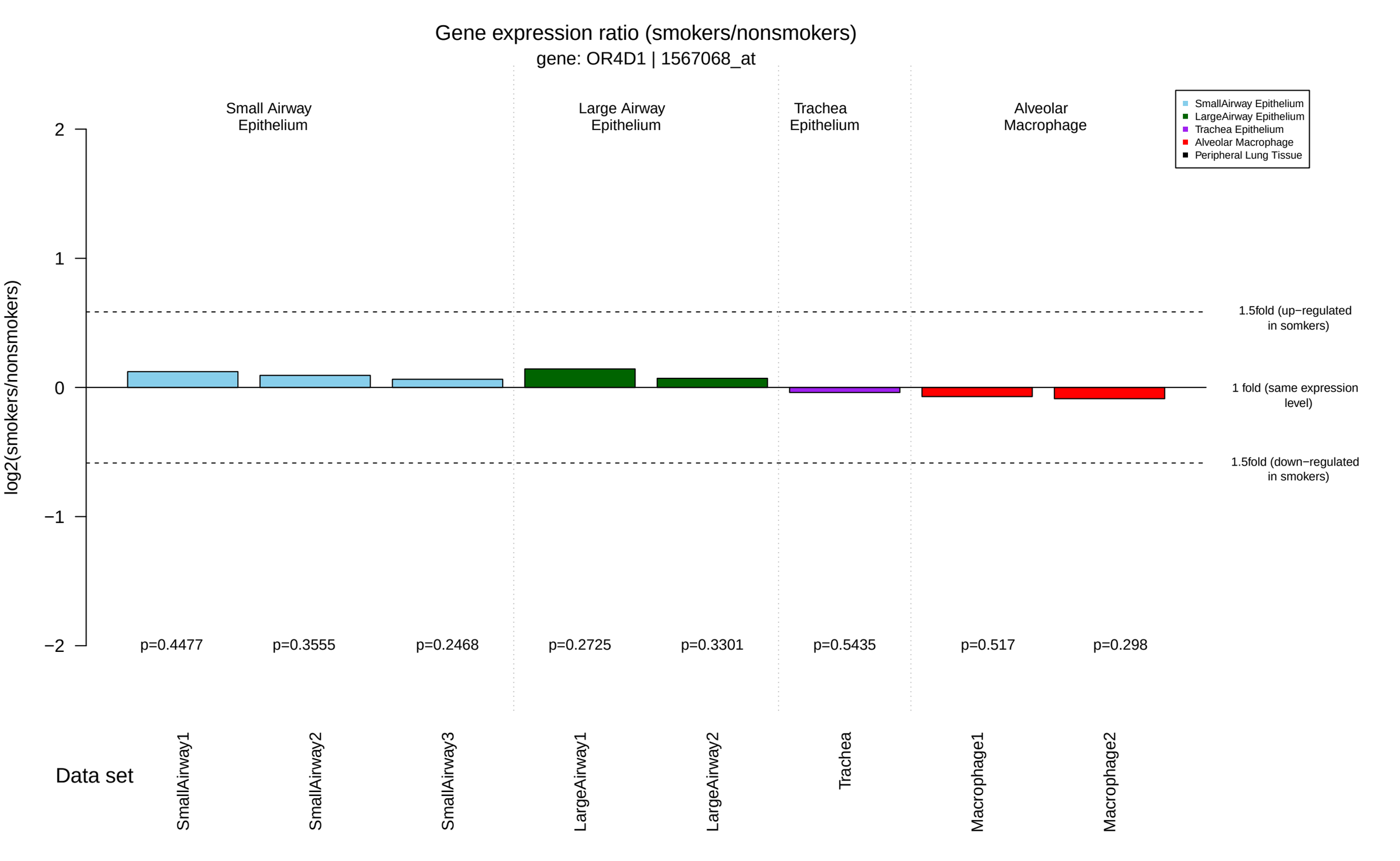
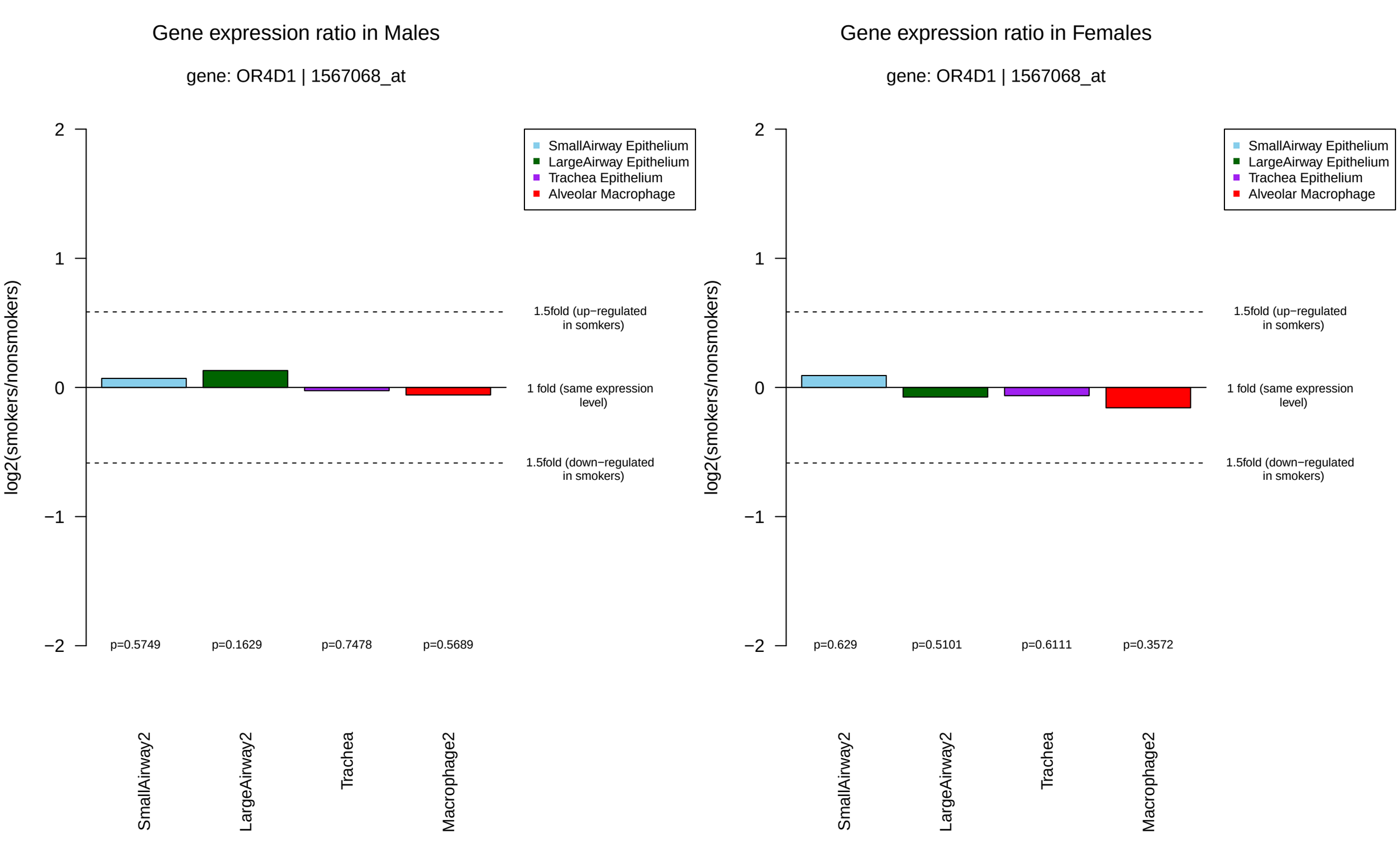
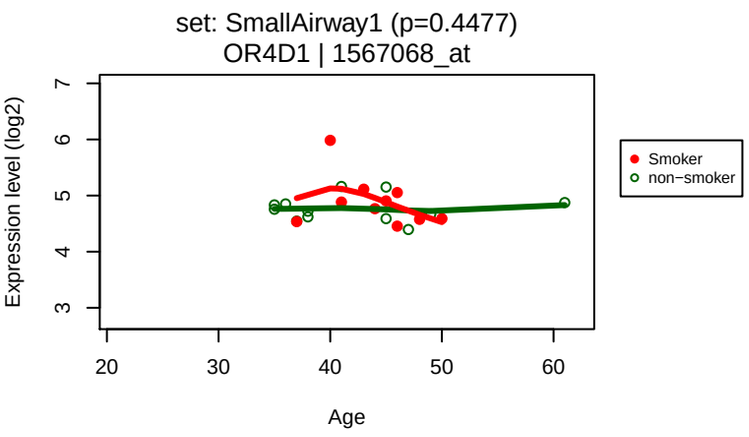
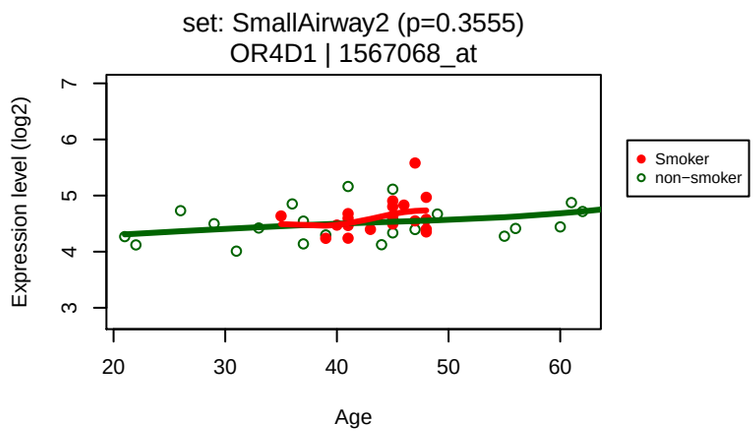
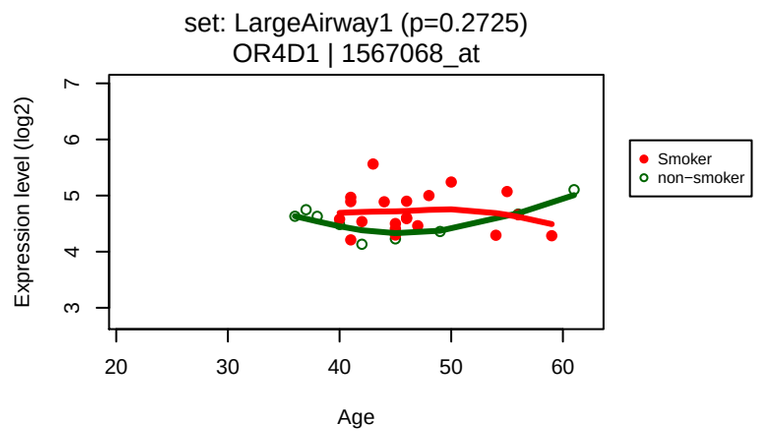

Gene expression of smokers and nonsmokers
gene: OR4D1 | 1567068_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4477
p=0.3555
p=0.2468
p=0.2725
p=0.3301
p=0.5435
p=0.517
p=0.298
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

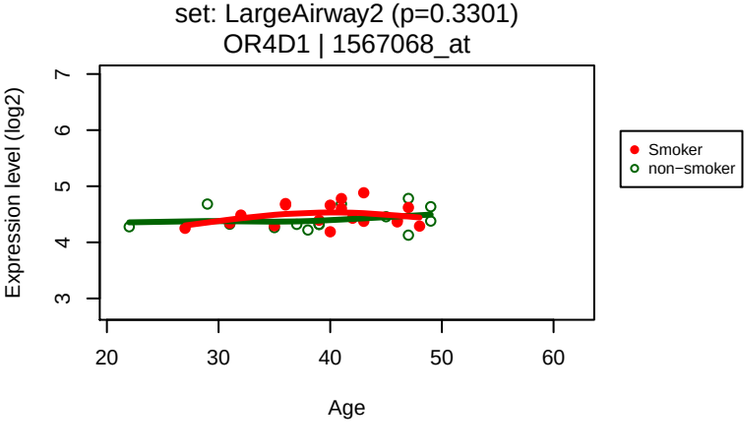

Gene expression ratio (smokers/nonsmokers)
gene: OR4D1 | 1567068_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4477
p=0.3555
p=0.2468
p=0.2725
p=0.3301
p=0.5435
p=0.517
p=0.298
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

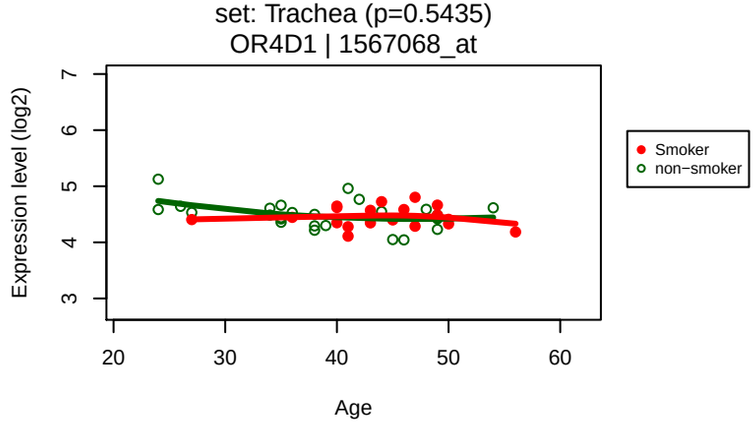

Gene expression ratio in Males
gene: OR4D1 | 1567068_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5749
p=0.1629
p=0.7478
p=0.5689
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR4D1 | 1567068_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.629
p=0.5101
p=0.6111
p=0.3572
SmallAirway2
LargeAirway2
Trachea
Macrophage2

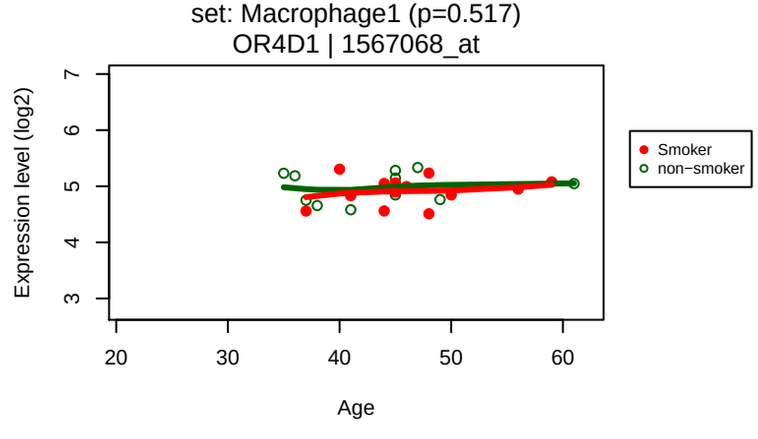

set: SmallAirway1 (p=0.4477)
OR4D1 | 1567068_at
Expression level (log2)
Age
Smoker
non−smoker

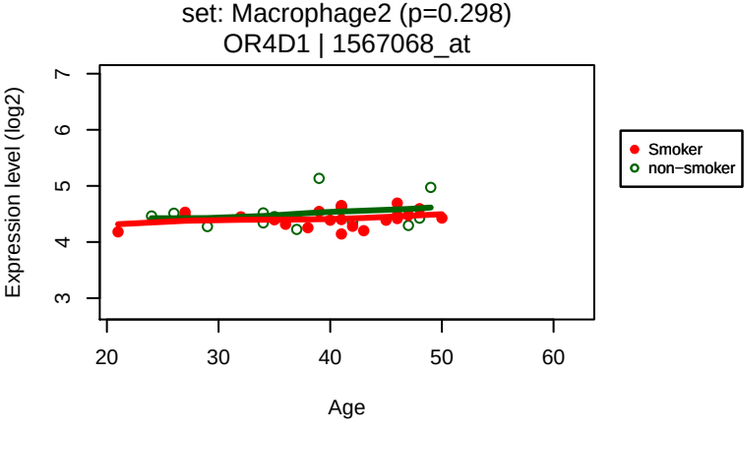

set: SmallAirway2 (p=0.3555)
OR4D1 | 1567068_at
Expression level (log2)
Age
Smoker
non−smoker

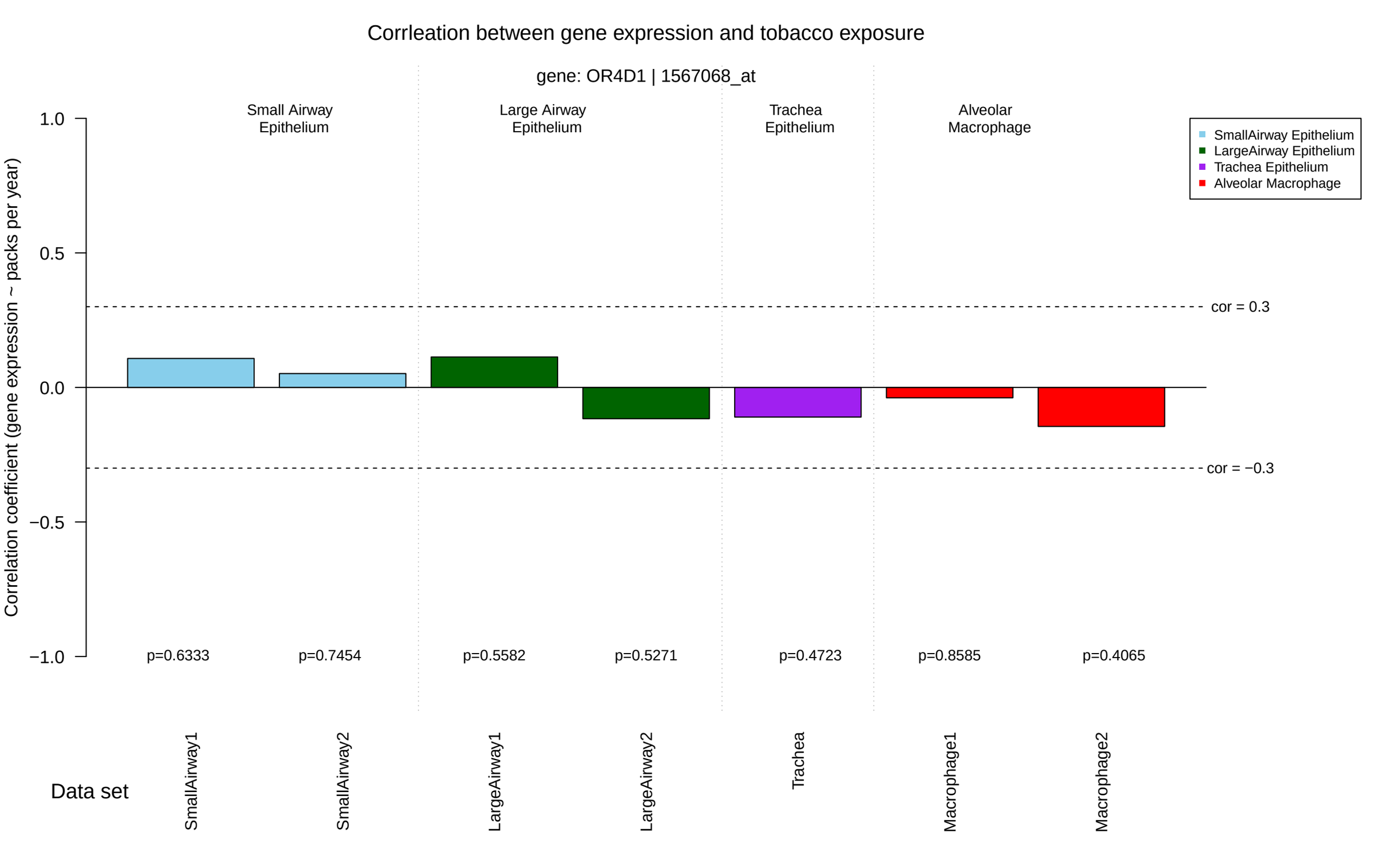

set: LargeAirway1 (p=0.2725)
OR4D1 | 1567068_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3301)
OR4D1 | 1567068_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5435)
OR4D1 | 1567068_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.517)
OR4D1 | 1567068_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.298)
OR4D1 | 1567068_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR4D1 | 1567068_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6333
p=0.7454
p=0.5582
p=0.5271
p=0.4723
p=0.8585
p=0.4065
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2